

3(a)	The straight line graph passes through the origin. This shows that the acceleration a of the particle is directly proportional to the displacement x from its equilibrium position. The straight line graph has a negative gradient. This shows that the acceleration a of the particle is opposite in direction to its displacement x, and directed towards the equilibrium position.	[1] [1]

(b)	Since $a = -\omega^2 x$, gradient $= -\omega^2$. From the graph, $\text{gradient} = -\frac{280}{4 \times 10^{-3}} = -70000$ Therefore, $\omega^2 = 70000$. $f = \frac{\omega}{2\pi} = \frac{\sqrt{70000}}{2\pi}$ $f \approx 42 \text{ Hz}$	[1] for gradient [1] for ω^2 link to gradient [1] for correct answer
(c)	Maximum speed, $v = \omega x_o = \sqrt{70000} \times 4 \times 10^{-3} \approx 1.1 \text{ m s}^{-1}$	[1] for substitution [1] for answer
(d)	Total energy $= \frac{1}{2} m \omega^2 x_o^2$ $= \frac{1}{2} \times 0.0050 \times 70000 \times (4 \times 10^{-3})^2$ $= 0.0028 \text{ J}$	[1] for correct substitution [1] for correct answer
4(a)(i)	$= 2.0 \times 10^8$	[1] for correct